

Decisions as Drivers of Value: Understanding Preference Change in College Students

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Abstract *(Written by: M. Thomas)*

The idea is often accepted that individuals hold internal preference states that remain relatively constant over time. However, psychological research demonstrates that the act of making decisions itself can influence the perceived importance of available options. In this study, a within-subjects design was employed to determine whether participants' ratings of the importance of three attributes of course selection (Instructor Quality, Grade Distribution, and Class Time) were altered by participation in a 20-trial forced-choice decision-making task. Participants ($N = 23$) completed pre- and post-task ratings of the importance of each attribute. Our hypothesis was that ratings of instructor quality would increase most significantly with completion of the decision-making task, compared to grade distribution and class time. The results were contrary to our expectations. Instructor quality ratings decreased post-task ($M = 4.22$ to 3.91 ; $t(22) = 1.82$, $p = 0.083$, $d = -0.39$), grade distribution ratings increased ($M = 4.09$ to 4.22 ; $t(22) = -2.03$, $p = 0.056$, $d = 0.43$), and class time did not change ($M = 3.48$ to 3.30 ; $t(22) = 0.44$, $p = 0.665$, $d = -0.09$). Since none of the differences achieved statistical significance, the lack of statistical power may be due to the small sample size at the end of the study (only 23 participants). As such, this study provides partial support for the concept of malleable preferences and supports the view that attribute diagnosticity plays a critical role in how decision-making tasks influence stated importance ratings.

Introduction *(Written by: M. Thomas; Edited by: S. Tang, A. Warner-Villarreal)*

An important question for the field of decision-making is how exactly individuals develop value for their options. From the perspective of classical economic theory, an individual's preferences should exist before the decision process and remain consistent regardless of the decision made. Therefore, according to this theory, the act of selecting one option over another is based on existing preferences and does not influence them. However, a large body of literature in both psychology and neuroeconomics suggests that this may not always be the case. Instead of having pre-existing preferences that guide our decisions, preferences may be constructed during the decision-making process itself. The development of these preferences is influenced by a wide variety of factors: the specifics of the decision-making task, the relative comparison of the alternatives, and the context in which the decision is made.

The idea that our preferences are derived from the decision-making process was first demonstrated through research regarding cognitive dissonance. Cognitive dissonance refers to the discomfort or distress an individual may experience when they must choose between two equally desirable alternatives. In one example, participants who committed to a choice rated the chosen option as more favorable and the alternative as less so.¹ These results do not support static theories that hold that our preferences remain constant throughout the decision-making process. Furthermore, this effect increased as the two options became more similar in terms of preference at the time of selection. When two options are similarly appealing to an individual, it creates greater conflict, thereby providing more opportunities for individuals to justify their ultimate decision.

Further evidence was produced against the assumption that one's preferences remain stable across various contexts and tasks, as stated by Slovic.² He proposed that our preferences often fluctuate widely due to differences in how questions are framed (task framing), how answers are elicited (response modes), and the options available. It remains an open empirical question whether changes in preference resulting from decision processes occur rapidly and systematically enough to affect long-term behavior. Research by Sharot et al. suggested that a change in preference resulting from the decision process could persist for extended periods.³ They demonstrated that changes in preference resulting from a decision process persisted for up to 1 year following the decision. Although they provided some insight into how decision processes can lead to long-term changes in an individual's subjectively valued options, additional research is needed to determine whether these effects occur consistently and reliably.

This study examines how quickly, through repeated trade-off decisions involving specific attributes, individuals' relative weighting of those characteristics can change. We have chosen to examine course selection as a decision environment because the college student population is naturally familiar with this type of decision, and there are many evaluative criteria for selecting courses. Research using surveys has shown that students' top priorities when choosing courses include instructor quality, class grade distributions, and scheduling.^{4, 5} From the decision-theory perspective presented in the course, instructor quality and positive grade distributions are examples of stimulus values. In contrast, class time is an example of an action cost; inconvenient class time does not diminish the intrinsic value of the course content but instead introduces additional effort or opportunity costs associated with enrollment.

We hypothesize that repeatedly making forced-choice decisions on the three course-selection attributes will increase the importance individuals assign to those attributes that best help them make decisions when faced with a multitude of conflicting trade-offs. The belief about this relationship matches previous research on value-based decision-making. That is, the ventromedial prefrontal cortex (vmPFC) has been shown to update the value signal to each option through repeated experience with trade-off decisions.⁶ The anterior cingulate cortex (ACC) measures the amount of "conflict" associated with difficult trade-offs.⁷ Both areas were thought to be responsible for changes in preference over trials. Specifically, we predict that Instructor Quality would have the largest increase in self-reported importance after task completion, followed by Grade Distribution, and that Class Time would either remain relatively unchanged or decrease slightly, as it is considered an action cost and thus has lower emotional salience.

Methods *(Written by: M. Thomas; Edited by: A. Warner-Villareal)*

Participants

We enrolled 23 college students in this study, recruited through UW-Madison's subject pool. From these, each provided complete pre- and post-task data and was included in the final analysis. Participants were limited to those currently enrolled full-time in college and who had registered for classes while attending university. Individuals who had

never used a university course registration system were excluded, as they would likely be unfamiliar with the course attribute labels used in the task (e.g., Rate My Professor score; MadGrades grade distributions).

Design

The experimental design for this study is a pre-post within-subjects design. The independent variable consists of three course attributes: Instructor Quality (the instructor's RateMyProfessor score), Grade Distribution (MadGrades), and Class Time. Two different types of data will serve as the dependent measures to assess how students' attitudes toward their courses are affected by these characteristics. First, students indicated on a scale of 1-5 how important they thought each of the three course attributes was to them. Second, we coded each trial as an aligned or reverse choice based on each participant's weighted preference score, as described below. Although there is no between-subject factor manipulated in our main analyses, we will explore whether individual difference factors moderate the relationship between students' course attributes and their choice behavior.

Procedure

The study was run as a web-based experiment accessible on any personal electronic device (laptop, tablet, smartphone). Participants completed all aspects of the session independently and remotely. After providing informed consent to participate in this research, participants first completed a rating-of-importance survey. In this survey, participants were asked to rate how important each of the three attributes used to compare courses (instructor quality, grade distribution, and class time) is to them when choosing which course to register for. The rating scale is a 5-point Likert-type scale anchored at 1 (not at all important) and 5 (extremely important). The overall "Rate My Professor" rating (1-5) measures instructor quality. Grade distribution is operationally defined as the percentage of students who received an A or B in the prior semester based upon data from "MadGrades." Class time is also operationally defined as the scheduled start time of the class (morning, noon, afternoon, or evening).

Participants then completed a 20-trial, forced-choice decision-making task. While completing the task, participants were shown two hypothetical courses side-by-side and were required to choose the one they would most like to enroll in. Both courses were fully described relative to all three attributes. The ratings for each attribute were randomly selected from a predefined set of stimulus materials, ensuring a consistent level of conflict across trials. Of the twenty trials, approximately ten included a clear majority opinion: one course had advantages over the other on at least two attributes. The remaining trials included more difficult choices in which both courses had advantages over the other on one or two attributes, so the participant had to give up one advantage to gain another. The order of trial presentation was randomized.

After participants completed the decision-making task, they were given 30 seconds of reflection time. The intent was to provide a time frame for the participant to assess and update preferences as they reflected on previously made decisions; it also served to reduce the influence of an immediate recency effect and to provide an opportunity for

the updated preference information acquired during the reflection period to become stable before subsequent assessments.

Reverse Choice Coding

Participant choice behavior during each trial was coded as either an “aligned” or a “reverse” choice, based on a weighted preference score derived from participants’ pre-trial ratings of attribute importance. The weighted preference score for each trial was determined through the summing of differences in rating across each of the three attributes that reflected each participant’s pre-trial ratings of importance. Choices made by a participant were coded as aligned when they selected the preferred option as indicated by their pre-trial ratings; otherwise, choices were coded as reversed. Then, for each participant, the reverse-choice proportion was computed as the ratio of reverse choices to the total number of trials (20). Finally, the association between participants’ post-preference changes and their patterns of choice behavior was explored. All data compilation and analysis were conducted in Google Sheets.

Results *(Written by: M. Thomas, A. Warner-Villarreal)*

Overview of Pre- and Post-Task Ratings

Table 1 below presents the mean importance ratings before and after completing the forced-choice task for three attributes used in course selection: Instructor Quality, Grade Distribution, and Class Time. Table 1 also includes the frequency of participants who rated an increase, a decrease, or no change in their ratings after completing the task.

Contrary to our predictions, Instructor Quality showed a decrease in mean importance ratings from pre- to post-task ($M_{pre} = 4.22$, $M_{post} = 3.91$) ($t(22) = 1.82$, $p = 0.083$, $d = -0.39$, 95% CI [-0.88, 0.06]). Grade distribution showed the only increase across the three attributes ($M_{pre} = 4.00$, $M_{post} = 4.22$) ($t(22) = -2.03$, $p = 0.056$, $d = 0.43$, 95% CI [-0.01, 0.55]). Class Time showed a slight decrease ($M_{pre} = 3.48$, $M_{post} = 3.30$) ($t(22) = 0.44$, $p = 0.665$, $d = -0.09$, 95% CI [-0.52, 0.34]). These results are presented in Table 1 and visualized at the individual level in Figure 1.

Table 1

Table 1 | Pre- and post-task mean importance ratings for course-selection attributes (N = 23). Values shown are means. Columns show the number of participants whose ratings increased, decreased, or remained unchanged.

Figure	N	Mean Pre	Mean Post	Increased	Decreased	No change
Instructor score	23	4.22	3.91	3	8	12
Grade score	23	4	4.22	8	3	12
Class time score	23	3.48	3.3	5	8	10
Early morning preference	23	2	2.48	6	1	16
Late morning preference	23	4.3	4.39	3	1	19
Midday preference	23	3.87	3.96	5	2	16
Afternoon preference	23	3.04	3.09	4	3	16

Individual Variability in Rating Change

Figure 1 shows the individual-level changes in ratings for each of the three attributes from pre-task to post-task. As shown for each of these three attributes, there appears to be considerable variation among participants relative to the general trend. Most participants displayed decreases or stable ratings post-task for both Instructor Quality and Class Time. However, a different picture emerges when examining the Grade Distribution: most participants showed increases in ratings after the task. Importantly, many participants for each attribute changed in the opposite direction from the overall mean for that attribute. This suggests that the decision-making process and/or the decision task did not affect all participants' self-reported preferences uniformly.

Figure 1

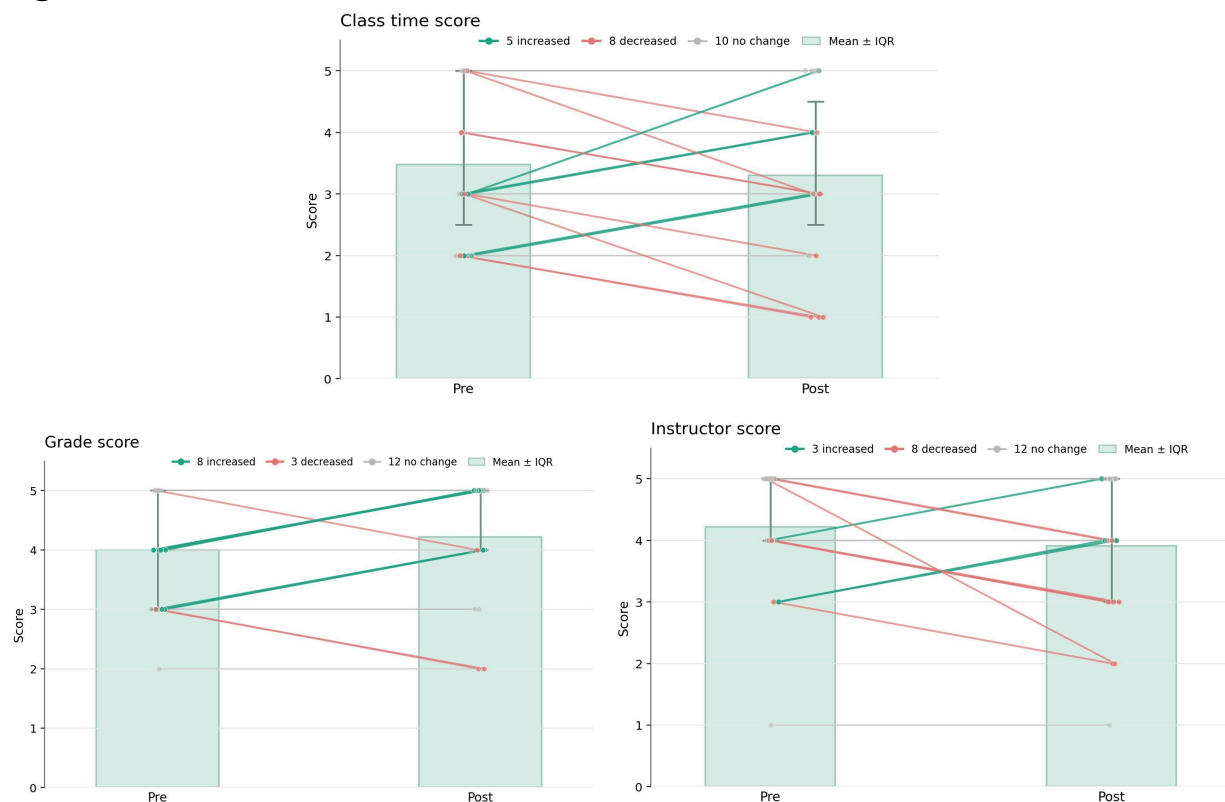


Figure 1 | Individual-level changes in Instructor Quality, Grade Distribution, and Class Time scores from pre-task to post-task across all participants. Green lines represent

participants whose ratings increased, red lines indicate decreases, and gray lines indicate no change. Shaded teal regions indicate the group means $\pm$ interquartile range (IQR) at each time point.

Choice Behavior Across Trials

Figure 2 illustrates the pattern of aligned and reverse choice behavior over all 20 trials for each participant. A very high proportion of participants' selections were aligned (weighted preference score ≥ 0) during the task. Thus, it appears that participants generally chose the course option that would have been preferred based upon the composite attribute weights assigned. Sporadic reverse choices occurred but did not appear to cluster around specific trials or participants. Therefore, it appears these reverse choices were due to trial-level decisional noise rather than any consistent derivation from preference.

Figure 2

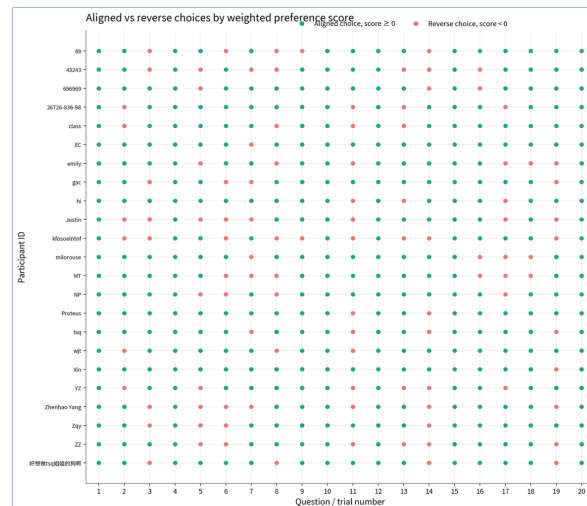


Figure 2 | Aligned and reverse choices by participant across 20 trials, coded by weighted preference score. Each row represents an individual participant, and each column represents a trial number. Green dots indicate aligned choices (weighted preference score ≥ 0) and red dots indicate reversed choices (score < 0).

Relationship Between Reverse Choices and Preference Change

The relationship between reverse-choice proportions and preference change was examined by computing Pearson correlation coefficients between participants' reverse-choice proportions and their relative pre-to-post importance rating changes for each attribute (Figure 3) to determine whether there was an association between decision-level behavior and post-task preference change. All three correlations were negative and small in magnitude: Instructor Quality ($r = -0.28$, $p = 0.199$), Grade Distribution ($r = -0.09$, $p = 0.674$), and Class Time ($r = -0.17$, $p = 0.429$). The widely

scattered data points are consistent with the weak effect sizes observed; together, these results suggest that reverse-choice proportion did not meaningfully predict the direction or magnitude of preference change for any attribute.

Figure 3

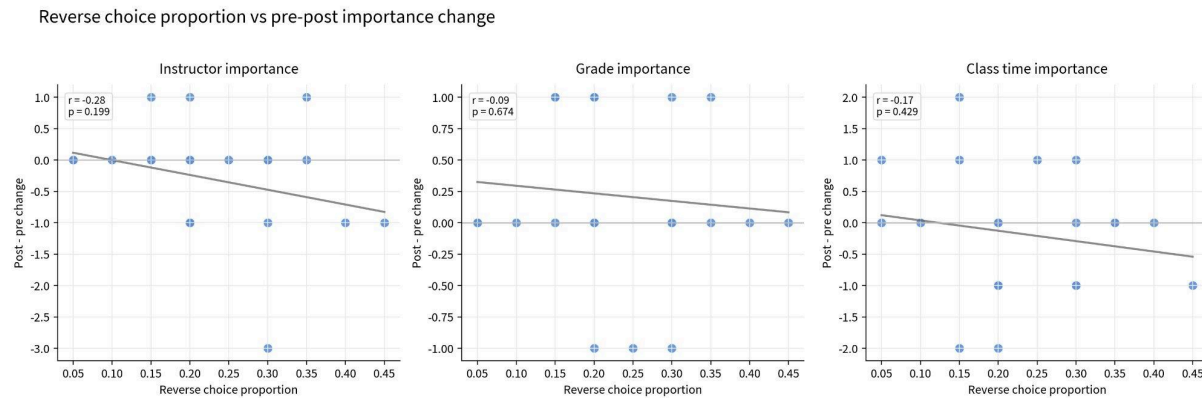


Figure 3. | Scatterplots of reverse choice proportion against pre-to-post importance rating change for Instructor Quality, Grade Distribution, and Class Time. Each panel displays a linear trend line and Pearson correlation coefficient (r). Instructor Quality: $r = -0.28$, $p = 0.199$; Grade Distribution: $r = -0.09$, $p = 0.674$; Class Time: $r = -0.17$, $p = 0.429$. None of the correlations reached statistical significance.

Discussion *(Written by: A. Warner-Villarreal; Edited by: M. Thomas)*

The findings only partially confirmed the hypotheses we had made. We predicted that Instructor Quality would show the largest increase after the task, followed by a moderate increase for Grade Distribution and a decline for Class Time. The contrary happened; Instructor Quality declined ($M = 4.22$ to 3.91 ; $t(22) = 1.82$, $p = 0.083$, $d = -0.39$), whereas Grade Distribution increased the most ($M = 4.00$ to 4.22 ; $t(22) = -2.03$, $p = 0.056$, $d = 0.43$) (Table 1). Class Time decreased somewhat ($M = 3.48$ to 3.30 ; $t(22) = 0.44$, $p = 0.665$, $d = -0.09$), consistent with our expectation that this action would be viewed as a cost and therefore discounted.

An alternative explanation is that Grade Distribution is a more tangible and direct measure of an outcome-relevant attribute compared to Instructor Quality. When repeatedly comparing course options across 20 trials, participants may have found Grade Distribution more decisive because of its direct connection to GPA and long-term academic standing. Instructor Quality, while rated highly at baseline, may have appeared less consequential when repeatedly traded off against concrete grade outcomes. The sample used in this study was also smaller than originally planned ($N = 23$), and some of the trends observed here may represent valid effects that a larger sample would find. This finding is also supported by research on preference construction, which indicates that the weight given to an attribute depends on both its original perceived importance and its relative importance across multiple trade-offs.²

Future work examining vmPFC activation could help clarify whether shifts in Grade Distribution importance genuinely reflect one's subjective importance updating, or whether it is simply ACC engagement during high-conflict trials that predicts the magnitude of the preference change.^{6, 7}

Reverse choice analysis failed to provide meaningful insight into the factors that cause individuals to alter their preferences among attributes. As shown in Figure 2, reverse choices were spread randomly throughout all data points, and as such, there was no statistically significant relationship between reverse-choice rates and the amount of change in an individual's preference for each attribute from before to after completing the survey (Figure 3). Therefore, it is likely that reverse choices simply reflected noise in the decision-making process rather than systematic construction resulting from the experimental procedure. However, one limitation of using self-report measures is that participants may not have sufficient information about the relative importance of the different attributes at any given point in time. One possible way to address this limitation would be to impose time constraints on respondents' responses per trial to determine whether and when they allocate more/less attention to certain aspects of the decision, depending on the specific trade-off being evaluated. The findings of this research contribute to a growing body of evidence suggesting that preferences are fluid and dynamically constructed over the course of decision-making, with relevant outcomes appearing to receive greater emphasis with more explicit, repeated trade-offs.

Neuroscience Extensions *(Written by: A. Warner-Villarreal; Edited by: M. Thomas)*

Connecting Behavioral Findings to Neural Mechanisms

The results of the current study, specifically the near-significant increase in Grade Distribution ratings ($t(22) = -2.03$, $p = 0.056$, $d = 0.43$) and the decrease in Instructor Quality ratings, demonstrate that repeated-forced choice decisions can produce meaningful shifts in stated attribute importance. The neural and computational mechanisms underlying this preference updating are discussed below.

Evidence Accumulation

To examine moment-to-moment evidence accumulation in future research, EEG or MEG can be employed to record the neural activity produced by individuals while making choices during the decision task. Both methods provide sufficient temporal resolution to detect the gradual buildup of decision-related signals throughout each individual trial.

The posterior parietal cortex (PPC) is expected to be the region primarily involved in evidence accumulation. Evidence from previous studies with both humans and non-human primates indicates that neurons within the PPC accumulate evidence linearly and continuously fire until their firing rate reaches a predetermined level of activation, signaling that a decision has been made.⁸ Kelly and O'Connell found an analogous signal in humans using EEG scalp recordings.⁹ They reported a lateralized,

slowly rising voltage that increased in amplitude in direct proportion to the amount and consistency of evidence accumulated across a trial. A similar relationship is predicted between the strength of this accumulation signal and the speed of responding. In trials where crystallization of the assigned weight to attributes is lacking, a shallower, less consistent accumulation signal was expected, along with longer response times. This pattern is consistent with drift-diffusion models of decision making that formally describe evidence accumulation as a noisy drift toward a decision boundary; therefore, it is also predicted that greater uncertainty in the assignment of attribute weights, as would occur early in the study, will result in slower, more variable responses.¹⁰

Subjective Value Encoding: fMRI

To understand how subjective values are encoded and updated, we would conduct a separate fMRI study in future work. Our principal goal is to determine whether BOLD activation in the ventromedial prefrontal cortex (vmPFC), previously shown to encode subjective values across various domains, corresponds to participants' pre- and post-task subjective assessments of the attributes.⁶ Specifically, if a participant rates a given attribute as more important post-task than pre-task, we expect to observe increased vmPFC activation in response to stimuli related to that attribute. Of particular interest is the near-statistically significant increase in ratings of Grade Distribution in this study. If such an effect were replicated in an fMRI environment, we would hypothesize that vmPFC activity would correlate with changes in the subjective value assigned to grade-related information throughout the task.

We plan to study the extent to which we can detect neural activity associated with conflict using the anterior cingulate cortex (ACC). The ACC has been shown to be active when an individual monitors for potential conflicts among options they may choose.⁷ Therefore, we expect that the degree of activation within the ACC should be greatest on those trials where the two options are most equally valued, and lowest on trials where one option is clearly dominant. Additionally, we anticipate that the degree to which individuals engage their ACC will directly relate to the magnitude of the shift in attribute importance after completing the task. In other words, we expect that participants who experienced great conflict, as indexed by higher ACC activation and longer reaction times on difficult trials, will show larger post-task shifts in attribute importance ratings, consistent with cognitive dissonance accounts of post-decisional preference change.¹ This prediction is supported by the present findings, in which Instructor Quality and Grade Distribution, the attributes most likely to generate trade-off conflict, showed the largest post-task rating shifts, while Class Time showed minimal change.

However, it is important to note that fMRI and EEG/MEG techniques represent complementary aspects of the decision-making process and therefore cannot be administered simultaneously to the same participants using conventional methods. For this reason, we suggest running them as two separate studies: one optimized for temporal resolution (EEG/MEG), and one for spatial resolution (fMRI). Together, they would provide a more comprehensive picture of the neural mechanisms underlying decision-driven changes in preference.

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Abstract

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Introduction

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Neuroscience Extensions

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References

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